

DHRUV GUPTA

Software Engineering | Data Science | ML Engineering

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SUMMARY

MS in Applied Data Science candidate at Indiana University with 3+ years of production experience building data pipelines processing data and ML systems achieving 92% accuracy. Optimized backend systems serving 10K+ daily users with 84% latency reduction, driving \$1M+ cumulative revenue impact. Co-founded tech consultancy delivering \$50K+ revenue across 10+ applications. Research Excellence Award recipient for LLM research with 85% query resolution. Strong foundations in PyTorch, distributed systems, SQL optimization, and cloud infrastructure.

SKILLS

- ML/AI: PyTorch, TensorFlow, Scikit-learn, CoreML, CNNs, ResNet, LLMs (PaLM 2, GPT), OpenAI Whisper, MLflow
- Languages: Python, SQL, JavaScript/TypeScript, Swift, R, Java, Kotlin
- Data Infrastructure: Snowflake, PostgreSQL, MongoDB, Redis, BigQuery, Elasticsearch, AWS Redshift, Neo4j, Firebase, Spark, Kafka, Airflow
- Cloud & DevOps: AWS (S3, EC2, Lambda), GCP, Docker, Kubernetes, CI/CD, Git, CUDA/GPU Computing
- Analytics & Tools: Pandas, NumPy, Tableau, Power BI, A/B Testing, ETL Pipelines, REST APIs, Microservices
- Others: Cross-functional Leadership, Agile/Scrum, Stakeholder Management, Client Relations (80% retention)

EDUCATION

Indiana University Bloomington

August 2025 - May 2027

MS in Applied Data Science

Bloomington, IN

- Upcoming: Deep Learning, Human-Robot Interaction, Engineering Cloud Computing
- Completed: Applied ML, Applied Database Technologies, Introduction to Statistics, Computer Vision, Information Visualization, Faculty Assistance in Data Science (FADS) Research

MIT World Peace University

August 2021 - May 2024

BSc in Computational Mathematics & Statistics

Pune, India

- Top of the of graduating class; Led multiple university-funded research and development projects
- Research Excellence Award: Architected personalized LLM on iOS using Google PaLM 2 with 85% query resolution rate
- Foundations: Convex optimization, Bayesian inference, probabilistic graphical models, numerical linear algebra

EXPERIENCE

Sallie Mae Corp.

May 2026 - Jul 2026

Software Engineering Intern

Indianapolis, Indiana

- Re-engineered Analytics infrastructure handling 1B+ events, reducing query execution times by 250% through Clustering and Micro-pruning and built a Streamlit dashboard for analyzing it, enabling near real-time business insights.
- Developed an automated ETL pipeline, shrinking 150+ attribute dataset to 20 business-critical fields, slashing compute costs by 70%. Built cloud-native data infrastructure using Python, Snowflake, AWS, and Azure DevOps to support enterprise-scale analytics.

Scientific Jugaadfunda Innovations LLP.

August 2023 - February 2024

Software Engineer - Data Systems

Pune, India

- Built data pipelines across MongoDB + PostgreSQL serving 50K+ daily users with 99.9% uptime
- Optimized SQL queries reducing latency 84% (2.5s to 400ms), enabling \$120K annual revenue increase
- Engineered real-time analytics dashboard tracking user behavior, increasing session duration 25% (12-15 min)
- Deployed microservices handling 10K+ concurrent requests with 65% latency reduction via Redis caching layer
- Delivered 4 data-driven applications serving 100K+ MAU with less than 0.1% crash rate

RootKit Consultancy Services LLP.

April 2023 - March 2024

Co-Founder & Technical Lead

Pune, India

- Generated \$50K+ revenue building 10+ data-driven iOS/Android apps with 10K+ downloads, 4.8-star rating
- Led 5-developer team achieving 80% client retention and 25% QoQ growth through ML-powered features
- Reduced costs 40% via lean operations while maintaining 100% on-time delivery and 60% profit margin

PROJECTS

Multi-Modal Computer Vision Pipeline

November 2025 - January 2026

PyTorch, ResNet-50, PCA, t-SNE, UMAP, K-Means, EM Clustering

- Architected end-to-end ML pipeline processing 40K+ images across 5 visual conditions achieving 92% classification accuracy
- Implemented dimensionality reduction preserving 95% variance with PCA, reducing features from 50K to 200 dimensions
- Deployed ResNet-50 transfer learning extracting 2048-dim features, improving clustering accuracy by 35%
- Built CNN classifier with 3.2M parameters achieving 0.08 validation loss in 50 epochs on GPU cluster